Provincial mock examination

DESIGN AND TECHNOLOGY DEPARTMENT

GRADE 9 **July 2024**

Paper 1 (Theory and Design) 608/1

***TIME: 2 HOURS* 30 MINUTES**

Time 2 hours 30 Minute **Total Marks: 155**

Instructions to candidates

1. Pull out the inserted **answer booklet** from the question paper.
2. Write your **examination number, school/centre** and **code** on the first page of the answer booklet.
3. Ensure that you write your **examination number** on **every page** of the answer booklet.
4. There are **three (3)** sections in this question paper; section A, B and C. Answer all question in section A and B.
5. Section C has three (3) parts;

   Part I has **one compulsory** question;

   Part II has **two** question, chose one question; and

   Part III has **two** questions, chose only one question.
6. Write **all** your answers in the answer booklet provided.
7. Use instruments and colour (enhancements) in section C part III (graphic communication)

Information for candidates

Cell phones are not allowed in the examination room.

SECTION A. [30 MARKS]
Multiple-choice questions

Answer **all** questions from this section. Write all your answers in the answer booklet provided. You are required to Circle or cross your best answer for section A

1 Which of the following is a correct safe working habit?

A. Cut metal with hacksaw

B. Smoothen wood with emery

C Use a mushroom head chisel correctly

D Do not run in the workshop.

2 If a chisel is used without a mallet, the action is referred to as

A Chopping

B Grooving

C Morticing

D Paring

3 Name the type of finish technique that is **NOT** applied on wood

A Blueing

B Staining

C Varnishing

D Painting

4 Aluminium is used for making aeroplane bodies because…

A Cannot conduct electricity

B Cannot conduct heat

C Is light in weight

D Has a good surface finish.

5 A business plan does the following except

A describing its goals and objectives

B describing the product to be produced and marketed

C describing the type of business to be undertaken

D identifying good and bad customers

6 In bricklaying ,which trade tool is used for checking the horizontal and vertical accuracy of brick or block work?

A cold chisel

B fuse

C spirit level

D trowel

7 What stage should systematic investigation to establish the facts be considered during design process?

A analysis

B brief

C research

D specification

8 Mwape has been buying white paper she cuts it properly to make envelopes which she sells.The cut paper with line in readiness for folding and gluing to make an envelopes is called….

A development

B kite

C plane paper

D trapezium

9 Tool steel is another name for…….

A high carbon steel

B high speed steel

C mild steel

D pig iron

10 Most handles of cooking utensils are made of thermosetting plastics because……..

A is a poor conductor of heat and electricity

B is hard ,strong and easy to handle

C resists corrosion and can easily be reshaped

D resists rust and it is colourful

11 Which of the following statement describes the best methods of stopping bleeding from wounds

A bandage right away with a clean cloth

B do nothing to it in order not to disturb the casualty

C Press the nearest pressure points ,then directly on the wound with cotton wool

D wash under running tap for thirty minutes or until bleeding tops

12 From the materials below, which one is a man-made?

A hardboard

B mukwa

C mutondo

D pine

Study the figure below and refer to it when answering question 13 and 14



Figure 6 shows a bottle top opener in use

Which one of the following is the pivot in this action?

A 1

B 2

C 3

D 4

14. Which class of leavers is shown in figure 6?

A First class

B Fourth class

C Second class

D Third class

15 If a corrosive liquid splashes in your eyes, you should …..

A ask a friend to blow them with air.

B close your eyes and let them rest

C wash them with plenty of running water

D Wipe them with a clean cloth

16 From the symbols shown below, select the one representing the diameter

A CL

B R

C Ø

D ©

17 The diagram in figure 2 shows four lines drawn at different angle to the horizontal plane

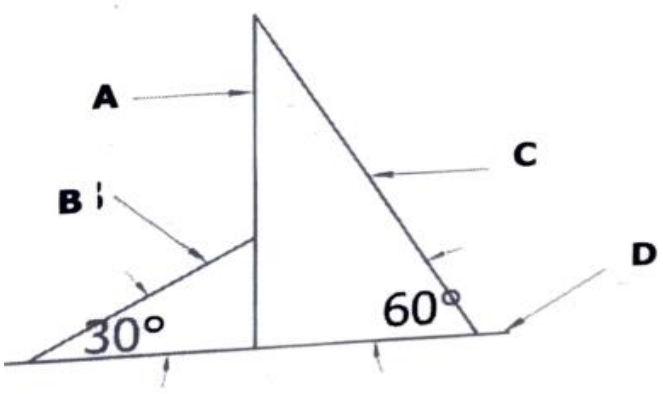
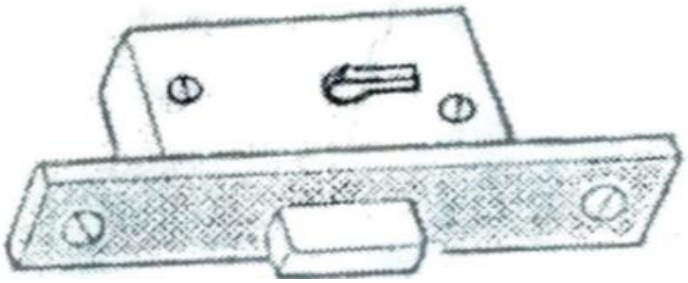


Figure 2

From the given lines A, B, C and D, choose a line that makes a right angle to the horizontal plane

18 The drawing shown below is a fitting in the house, name the drawing.



A hinge

B latch

C mortise lock

D pad lock

19 Why do we reduce moisture in timber ? to…….

A improve its appearance

B make it hard

C make it soft

D make it suitable to work with

20 Storage area of plastics should be……..

A damp and hot

B dust free and non-greasy

C exposed to sunlight

D filled with water

21 Which one of the following is NOT a characteristics of an Entrepreneur?

A look for opportunities

B report to the manager early

C seek information

D take initiative

22 Which of the following is the correct order of making a shape from any material ?

A cut, measure, mark, sand

B mark, measure, cut, sand

C measure, mark, cut, sand

D sand, cut, measure, mark

23 What alloy is obtained from the mixture of copper and zinc ?

A brass

B bronze

C soft solder

D stainless steel

24 The following are manufactured board except……

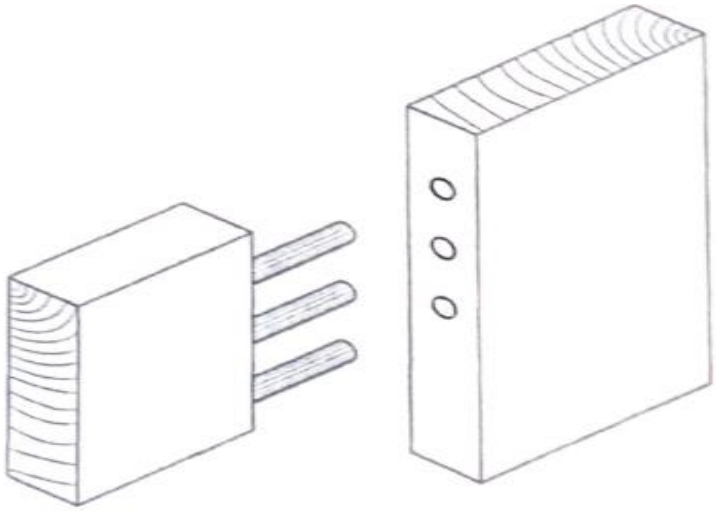
A block board

B chip board

C medium density fiberboard

D urea formaldehyde

25 Diagram below shows one of the joints used in place of mortice and tenon joint .use it to answer question 25 and 26



The joint above is called……..

A butt joint

B cross halving joint

C dowelled joint

D tee halving joint

26 Where is the joint mostly applied ?

A box construction ,cabinets and drawers

B chair, frame and stool construction

C concealing the end grains of the piece of wood

D joining timber which are at right angles to each other

27 ……is an example of a shell structure

A bridge

B pot

C table

D tree

28 Which adhesive is commonly used when joining wood to wood

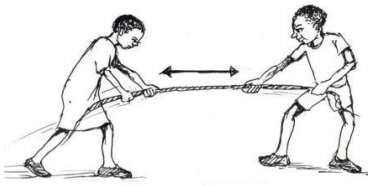
A casein glue

B contact glue

C P V A glue

D resin glue

29. State the type of force illustrated by the arrows shown in figure below.



A Bending force

B Compression force

C Tension force

D Torsion force

30. **Figure 6** below shows two PVC pipes to be joined together at **K**. which of the following glues would be appropriate to use?

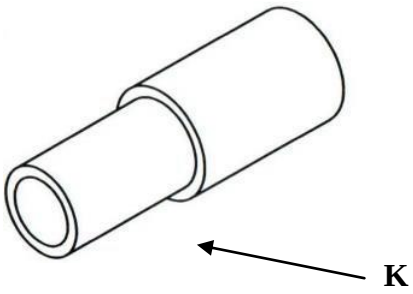


Figure 6

A. CPVC adhesive.

B. Epoxy adhesive.

C. PVC adhesive.

D. Synthetic wood adhesive.

Section B [30marks]

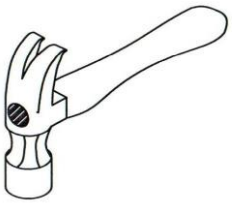
Answer all questions

Write your answers in spaces provided in the ***answer booklet.***

31. a) The figure below shows carpentry tools. Name tools and state the use of each tool.



Y



Z

Figure 7

Tool Y: _____ use : _____

Tool Z: _____ use: _____ [4]

32. Figure 2 below shows one of the **three** classes of levers.

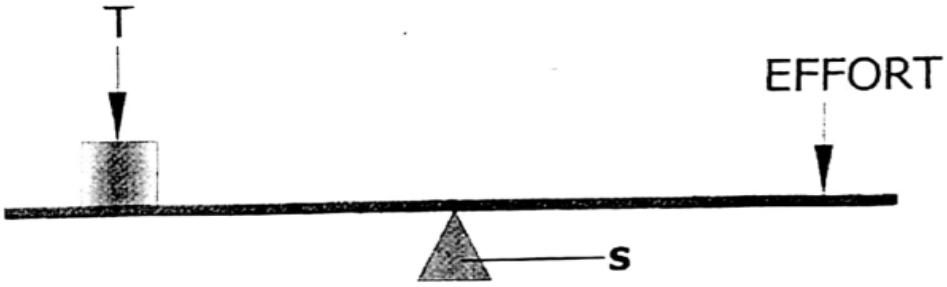


Figure 8

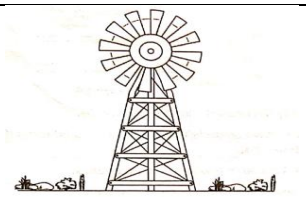
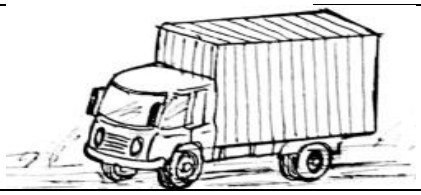
(a) To which class of levers does **figure 8** belong?_____[1]

(ii) Write the names of the parts labeled **S** and **T.**

S _____ [1]

T _____ [1]

33. Complete the table below by filling in the correct information. [3]

	system	Energy source
(i)		
(ii)		
(iii)		

34. a) Suggest the best finish you would apply on the following artifacts;

i) Wooden chopping board._____ [1]

ii) Key steel door frame _____ [1]

iii) Mention at least **two** qualities of a good entrepreneur

_____ [1]

_____ [1]

b) i) Name one types of lighting and ventilation and give example each. [4]

Type of lighting : _____ example: _____

Type ventilation : _____ example: _____

ii) Give two reasons why first aid must be given to a victim. [2]

1st reason: _____

2nd reason: _____

iii) Outline three processes that are involved in timber processing starting from a standing tree in the forest up to the stage where timber is for use. [3]

1st process: _____

2nd process:_____

3rd process: _____

iii) Sketch the warning sign to alert people that in that place eating food is not allowed. [5]

vi) Write down two effects of heat on plastics.

_____

_____ [2]

SECTION C [90 marks]

This section has three parts

Read and follow instructions in each part.

Part I - Compulsory

35. The situation in **Figure 10** is obtaining in Mwimba Kennedy's bedroom. Study the situation and answer the questions that follow

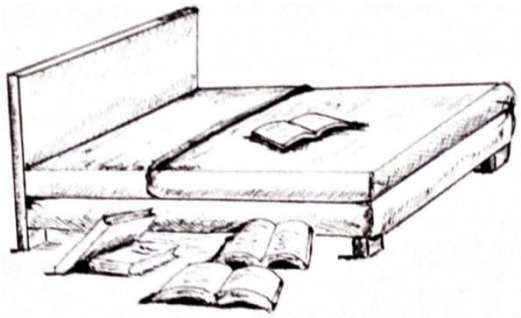


Figure 10

i) Identify the problem from the situation in **figure 10**.
[3]

ii) Write down a design brief from the problem in **(i)** above. [4]

iii) **S**ketch **two** possible solutions in pictorial view for the situation above. [18]

iv) List down **two** safety measures you have put in consideration in your design for the solution in **'(iii)'** above [2]

iv) List **four** specifications that can be considered when coming up with your design solution. [4]

v) Suggest **two** materials you would need to make solution you have sketched **'(iii)'** above. [2]

vi) Give **one** reason for each material you have suggested. [2]

[Total 35marks]

Part II

There two questions in this part. Choose **one** question.

36. Either (a) Plastics are non-metallic materials that are used to make various products on the market.

i) Write down two uses of plastics. [2]

ii). Write down two safe way of disposing plastics [2]

iii). Name and sketch two main methods of converting timber. [8]

iv) **Figure 11** shows a young woman drilling the hole using an electric drilling machine.

Identify the two parts of the body she is protecting and the type of protective attire she has used. [2]



Figure 11

v) Write down three safety rules you need to observe when you are working in the workshop. [3]

vi) Name **four** items among others that must be found in the First Aid box [4]

vii) Name and Sketch three methods of nailing. [9]

Total [30 marks]

Or **b)** **i)** Write down one property of softwood and one property of hardwood. [2]

ii) Name and sketch two manufactured boards. [6]

iii) Name and sketch one tool used in bricklaying [4]

vi) Draw the cross section of the tree and label five parts. [10]

v) The **figure 10** below shows a cabinet for storing tools in the workshop. What two materials would you recommend to use to make the cabinet? [2]

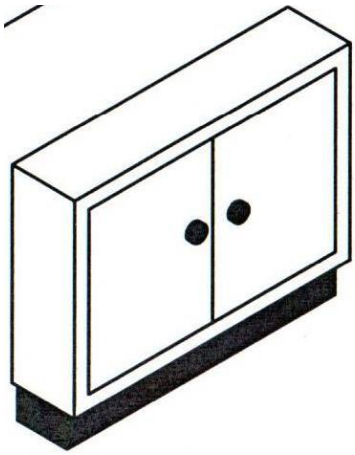


Figure 12

vi) Suggest two materials that can be used to make a handle [2]

vii) Suggest the method you can use to fix the handle onto the doors. [2]

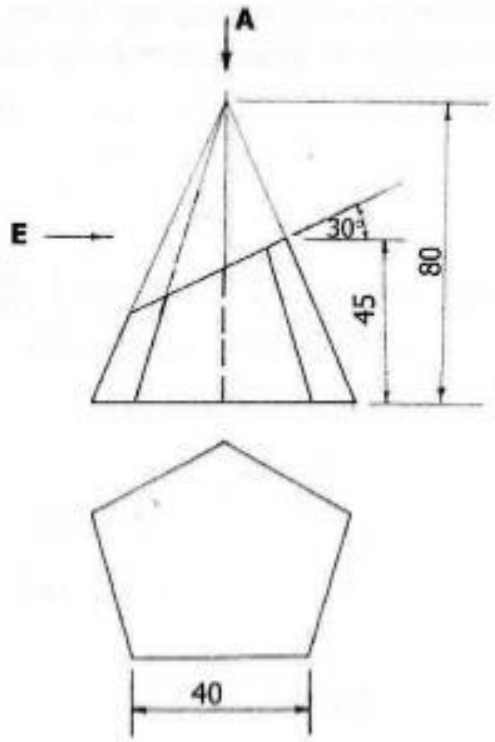
viii) Write two reasons why water pipes today are made using plastic instead of metal. [2]

Total [30 marks]

Part III

There two questions in this part. Answer **one** question.

37. either a)



i Draw the complete plan.
ii Draw the front elevation.
iii Draw the surface development of the rectangular base only seam along 30° cut.

Total [30 marks]

Or b) A swimming pool has just been constructed at a new school. The school has has asked you to design some signs that should graphically give the following messages:

i) Do not push others in the pool. [15]

ii) Do not smoke [15]

Total [30 marks]